

Probability and Statistics: Assignment II

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Instructions:

1. This is the second group assignment of STAT8225.
 2. Solve it in groups of 6 students in each group.
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1. A lottery selects one number uniformly at random from the integers 1 to 20.
 - (a) What is the probability that the selected number is divisible by 5?
 - (b) What is the probability that the number lies between 8 and 15 (inclusive)?
 - (c) Find the expected value of the selected number.
2. A cloud service company monitors user account retention. Based on system analytics, it is estimated that for users of the same subscription tier, the probability that a user account will still be active after 30 months is $\frac{2}{3}$. This probability is obtained from historical system usage data. If a sample of 5 new users is registered in the system, answer the following questions:
 - (a) What is the appropriate probability distribution to be applied here?
 - (b) What is the expected number of user accounts that will still be active after 30 months?
 - (c) Calculate the standard deviation of the distribution.
 - (d) Find the probability that at least three user accounts are still active after 30 months.
3. The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that
 - (a) at least 10 survive
 - (b) from 3 to 8 survive
 - (c) exactly 5 survive?
4. Find the probability that in a family of 4 children there will be
 - (a) at least 1 boy,
 - (b) at most 1 boy.

5. Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day. What is the probability that on a given day tankers have to be turned away?
6. A call center receives on average 6 calls per minute. Assume that the number of calls follows a Poisson distribution.
 - (a) What is the probability that exactly 4 calls are received in one minute?
 - (b) What is the probability that at least 2 calls are received in one minute?
 - (c) What is the probability that no calls are received in one minute?
7. The waiting time (in minutes) for a bus is uniformly distributed between 0 and 15 minutes.
 - (a) What is the probability that a passenger waits less than 5 minutes?
 - (b) What is the probability that the waiting time is between 7 and 12 minutes?
 - (c) Find the expected waiting time.
8. The lifetimes of a certain type of battery are normally distributed with mean 200 hours and standard deviation 25 hours.
 - (a) What is the probability that the batteries last more than 220 hours?
 - (b) What is the probability that the batteries last between 150 and 210 hours?
 - (c) Find the lifetime below which 10% of the batteries fall.
9. Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 527 and a standard deviation of 112.
 - (a) What is the probability of an individual scoring above 500 on the GMAT?
 - (b) How high must an individual score on the GMAT in order to score in the highest 5%?
10. A data analyst in a tech company is studying the relationship between the number of hours spent on system maintenance (X) and the system downtime in hours (Y) recorded over 12 days. The observed data are given below:

X	1	2	3	4	5	6	7	8	9	10
Y	2	3	5	4	6	7	8	9	10	12

- (a) Determine the least squares regression line of the form

$$Y = a + bX$$

that best fits the data.

- (b) Use the regression equation to predict the system downtime when maintenance time is 7 hours.
- (c) Compute the Pearson correlation coefficient between X and Y .
- (d) Interpret the value of the correlation coefficient in the context of system performance.
- (e) Comment on whether the linear model is appropriate for this dataset.

11. The average monthly electricity bill in a town is claimed to be 30,000 RWF. A sample of 10 households gives a sample mean of $\bar{x} = 32,000$ RWF and sample standard deviation $s = 4,000$ RWF.

Task: Test the claim at the 5% significance level.

12. A manufacturer claims that light bulbs last 1000 hours on average. A sample of 49 bulbs has a mean lifetime of $\bar{x} = 980$ hours and a known population standard deviation $\sigma = 70$ hours.

Task: Test the claim at the 5% significance level.